PALEOMAGNETISM OF THE TRIASSIC NICOLA VOLCANICS AND GEOTECTONICS OF THE QUESNELLIA SUBTERRANE OF TERRANE I, BRITISH COLUMBIA

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ABSTRACT

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The Triassic Nicola Volcanics are the oldest strata in the Quesnellia Subterrane of composite Terrane I which is located in the southcentral Cordillera of British Columbia. A paleomagnetic study using AF and thermal demagnetization methods of 563 speciments from 56 sites in three petrologically and tectonically distinct belts of the Nicola isolates two remanence components residing dominantly in hematite. The prefolding A component found after 560°C thermal cleaning gives a pole position of 11.3°W, 52.7°N ($\delta_p = 4.7°$, $\delta_m = 7.2°$), and appears to be caused by alteration associated with major Early Jurassic intrusive events. The dominant postfolding B component is best defined by 565°C thermal cleaning and gives a pole of 303.0°W, 82.5°N ($\delta_p = 6.6°$, $\delta_m = 8.6°$). It likely reflects alteration associated with Eocene volcanism. Combined with previous paleomagnetic data, these results indicate that Quesnellia was translated northward by $\sim 13°$ of paleoatitude along with the rest of Terrane I and rotated $\sim 45°$ clockwise during Late Cretaceous to Oligocene time with $\sim 7°$ of the translation and $\sim 15°$ of the rotation occurring during Eocene to Oligocene time.

INTRODUCTION

The Upper Triassic Nicola Volcanics are the oldest extrusive rocks exposed in the Quesnellia Subterrane of allochthonous composite Terrane I (Monger *et al.*, 1982) in southcentral British Columbia (Fig. 1). Paleomagnetic studies on Lower Jurassic plutons have shown that Quesnellia has been translated northward by $\sim 13°$ of paleolatitude and rotated clockwise by $\sim 43°$ between Late Cretaceous and pre-Miocene time (Symons, 1973, 1983a) as part of the overall motion of Terrane I (Monger and Irving, 1980; Symons, 1983b) and of the Cordillera overall (e.g., Beck *et al.*, 1981). Recent detailed mapping of the Nicola Volcanics by Preto (1979) provides the geologic basis for extending the geotectonic study of Terrane I.

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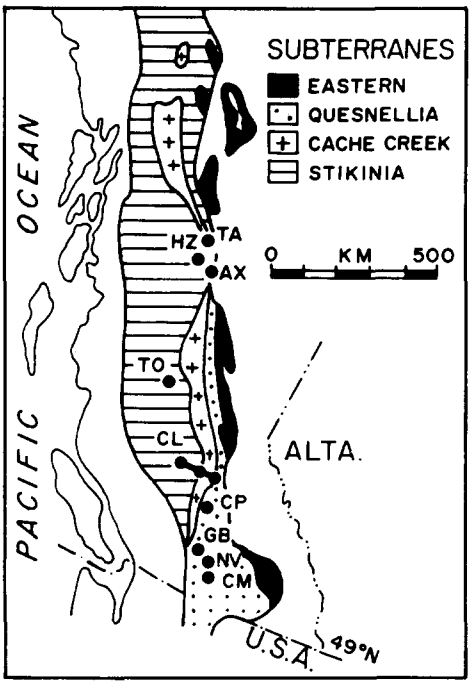


Fig. 1. Subterranes of Terrane I in British Columbia and southern Yukon of the Canadian Cordillera after Monger *et al.* (1982). Paleomagnetic data locations include the Takla Volcanics (TA), Hazelton Volcanics (HZ), Axelgold Intrusion (AX), Topley Intrusions (TO), Cariboo Lavas (CL), Cariboo Plugs (CP), Guichon Batholith (GB), Nicola Volcanics (NV) and Copper Mountain Intrusions (CM).

GEOLOGY

The Nicola Group is exposed in three north-trending belts bounded by major faults (Fig. 2). The Central Belt was deposited first and it is comprised of subaerial and submarine andesites and basalts, breccias, conglomerates and lahars with minor comagmatic diorite and syenite intrusions. The Eastern Belt consists of volcanic sedimentary rocks, lahars with minor basalt flows and syenite stocks. The Western Belt was deposited last and it is comprised of andesitic to rhyolitic flows and pyroclastics with interbeds of limestone, volcanic conglomerate and sandstone. The Central and Eastern Belts contain alkalic and calc-alkalic rocks derived from comagmatic alkalic intrusions whereas the Western Belt is calc-alkaline and derived from external sources (Preto, 1979). The volcanics are mostly massive and unfoliated, and they have been metamorphosed to the prehnite-pumpellyite grade. Late Karnian to Middle Norian fossils indicate a 218 ± 4 Ma age for primary deposition on the Odin (1982) time scale. This

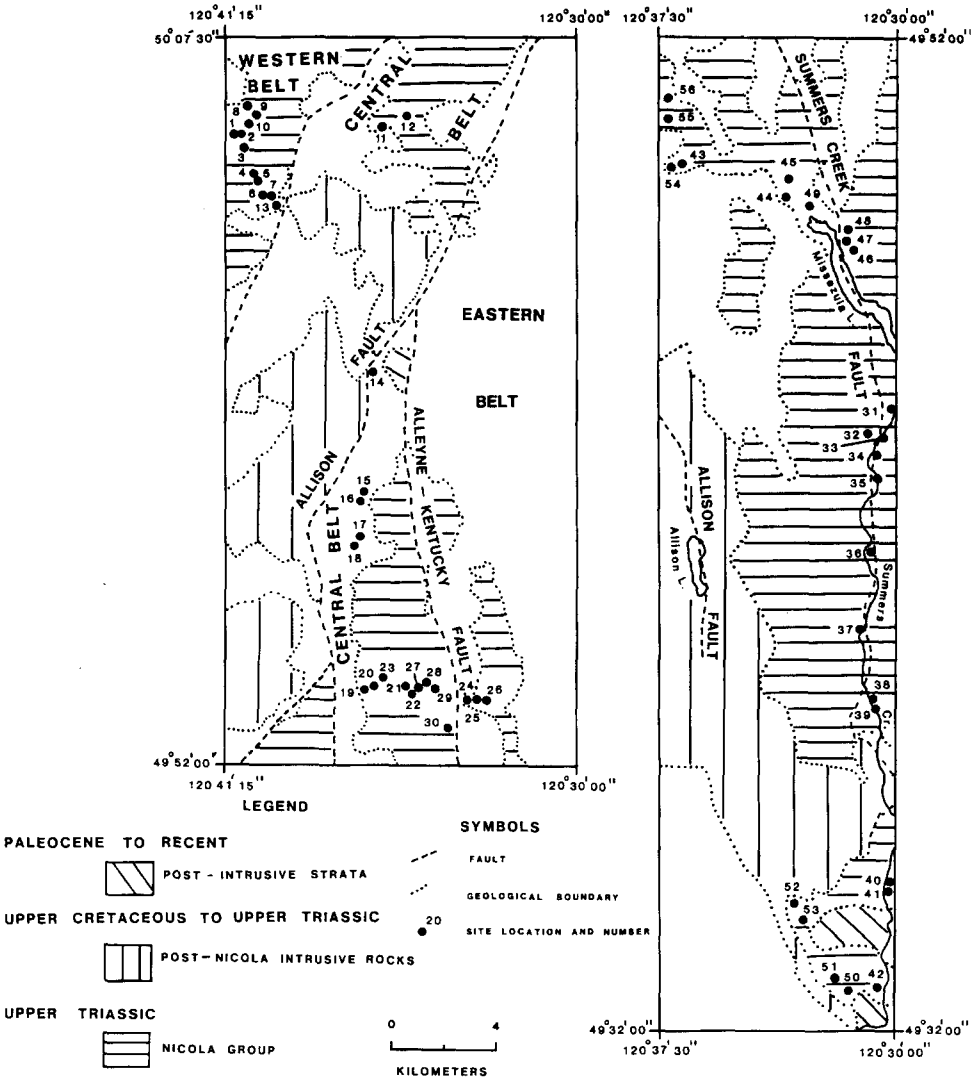


Fig. 2. Geology of the Nicola Volcanics in the sampling area after Preto (1979) showing site locations.

age is consistent with minimum K-Ar muscovite and biotite ages of 207 ± 5 and 204 ± 5 Ma (recalculated using the Steiger and Jäger (1977) constant) on the crosscutting Allison Lake granitic pluton. Several Lower to Middle Jurassic and younger stratified units rest unconformably above the Nicola Group. The Alleyene-Summers Creek and Allison fault systems define the east and west sides of the belts, respectively. These systems are thought to be part of an ancient rift that has been active since the Early Triassic and

that has controlled the extent and distribution of the Nicola and younger rock units. The Nicola rocks are not folded, but rather they are simply tilted in a series of monoclinal panels that were formed for the most part by Early Jurassic time (Petro, 1979).

METHODS

The 1:15840 scale geologic maps of Preto (1974, 1975 & 1976) were used to locate 56 sites with 34, 11 and 11 from the Central , Eastern and Western Belts respectively (Table 1, Fig. 1). Except for one gabbro site (#29), all are from massive extrusive units. Their structural attitudes are well defined by flow tops or interbedded sediments. Five or six cores from one flow were collected from each of the 47 drilled sites and three hand samples were collected from each of the remaining nine sites. From seven to 15 specimens were prepared for each site giving a total of 563 specimens.

After measuring their natural remanent magnetization (NRM), one typical specimen from each site was alternating field (AF) demagnetized in 11 steps up to 100 mT and a second typical specimen was thermally demagnetized in 9 steps to 650°C using Schonsted SSM-1A and TSD-1 instrumentation. After examining this step demagnetization data, all remaining specimens were first AF cleaned at 30 mT and then half of the specimens were thermally cleaned at 565 and 650°C. Accepted specimen vector directions were required to have a within-specimen angular standard deviation (ASD) of $< 20°$ (Harrison, 1980; Briden and Arthur, 1981; Stupavsky and Symons, 1982). Conventional tiered statistics (Fisher, 1953) and density contouring modal analysis (Kamb, 1959; Van Alstine, 1980; Stupavsky and Symons, 1982) were used to analysze the remanence directions both before and after bedding tilt correction for each tectonic belt in turn and for the total collection.

AF RESULTS

The NRM intensities fall in the 10^{-3} to 10^{1} Am^{-1} range. Uncorrected for bedding tilt, the NRM directions for all three belts give the same single modal anomaly direction of high statistical significance ($> > 99.95\%$ confidence) that is close to the present Earth's magnetic field (EMF) direction of $D = 22.7°$, $I = 72.4°$ (declination, inclination) (Fig. 3a). Tectonic correction splits each mode into two populations that reflect the dominantly easterly and westerly intermediate dips of the flows. The post-deformational origin of the NRM is also sugested by the fact that $< 10\%$ of the vectors are reversed or negatively inclined which may imply significant viscous remanent magnetization (VRM) components.

TABLE I

Site geologic and sampling data

Site	Belt	Unit	Attitude		Samples		Site	Belt	Unit	Attitude		Samples	
			Strike °	Dip °	N	n				Strike °	Dip °	N	n
1	W	3a	25	60	5	9	29	C	5	355	70	5	9
2	W	3a	25	60	5	9	30	C	1c	340	50	5	7
3	W	3a	20	55	5	7	31	E	2a	170	45	3	10
4	W	3b	45	75	5	8	32	F	2a	160	45	3	10
5	W	3a	40	70	5	8	33	E	2a	180	40	3	12
6	W	3a	30	65	5	8	34	E	2a	180	40	3	12
7	W	3d	30	70	5	9	35	E	2a	170	50	5	9
8	W	3a	20	60	5	9	36	C	1a	5	50	5	9
9	W	3a	25	55	5	9	37	C	1a	180	60	3	7
10	W	3a	35	55	11	38	E	2c	225	70	5	9	
11	C	1a	30	55	5	9	39	E	2c	180	85	3	10
12	C	1a	40	40	5	6	40	C	1e	15	68	3	14
13	W	3a	35	70	5	9	41	C	1a	355	53	3	10
14	C	1a	200	65	5	9	42	C	1e	230	60	3	8
15	C	1e	110	55	5	9	43	C	1b	0	25	6	11
16	C	1e	110	60	5	9	44	C	1a	350	50	5	9
17	C	1a	175	70	5	9	45	C	1a	340	70	5	8
18	C	1a	175	70	5	9	46	E	2a	355	50	5	9
19	C	1e	165	85	5	8	47	E	2b	170	35	5	8
20	C	1c	175	85	5	9	48	C	1a	160	60	5	9
21	C	1c	345	60	5	9	49	C	1a	340	60	5	10
22	C	1c	340	50	5	8	50	C	1e	168	56	5	9
23	C	1c	350	70	5	9	51	C	1a	210	85	5	9
24	E	2a	170	60	5	9	52	C	1a	50	45	5	9
25	E	2a	170	60	5	9	53	C	1a	65	34	6	10
26	E	2b	170	60	5	9	54	C	1a	0	25	5	7
27	C	1c	355	73	5	9	55	C	1c	0	25	5	9
28	C	1c	345	65	5	9	56	C	1a	10	30	5	9

Notes: W, C and E are the Western, Central, and Eastern Belts, respectively.
Unit designations follow Preto (1979). N and n represent the number of cores or samples and specimens analysed.

The remanence resists AF step demagnetization. The median of the 56 median destructive fields is ~ 17.5 mT with 13 values of > 100 mT. At 100 mT the median of the ratio of measured to initial NRM intensity (Jn/Jo) is ~ 0.15 with 13 values of > 0.50. The low median value of ~ 15 deg/dT (mdeg/oe) for the paleomagnetic stability index (PSI) (Symons and Stupavsky, 1974) with only five values of > 100 indicates that the

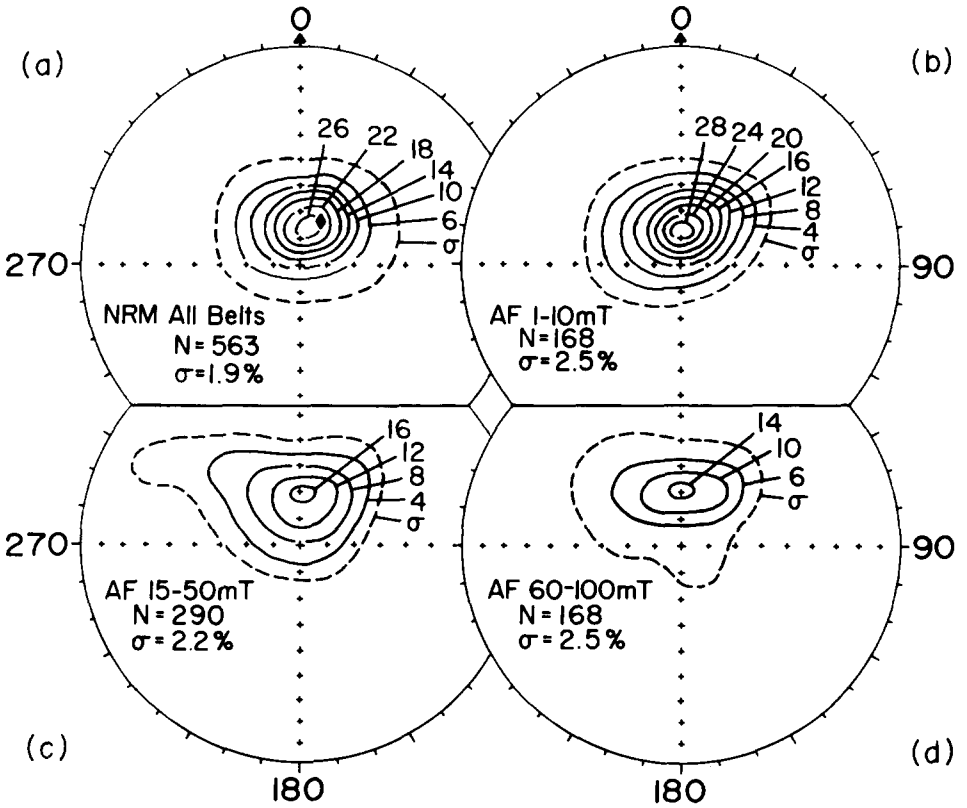


Fig. 3. Equal-area stereonets showing the population density contours for vectors from all tectonic belts combined without tilt correction showing: a) NRM; b) AF step from 0 to 10 mT; c) AF step from 15 to 50 mT; and d) AF step from 60 to 100 mT. The conventions are as follows: density contours in percent; δ is the 95% confidence level; and N is the number of vectors in the population.

remanence directions are highly stable and unchanged by AF step demagnetization. The median of the fields for the optimum PSI values occurs at ~30 mT and so this value was used for AF demagnetization of all specimens. The measured directions were next divided into 0 to 15, 20 to 50 and 60 to 100 mT groups for plotting and analysis. When tectonically uncorrected directions from all belts are combined (Fig. 3b, 3c, d), the location of the mode is virtually unchanged with increasing field although its peak density decreases progressively from $>28\%$ to $>16\%$ to $>12\%$ with the latter value still being $>>99.95\%$ significant. When each belt is examined independently, they give the same mean direction and peak density decay pattern, Tectonic correction yields two or more modes for each belt, and for the collection as a whole, confirming that the remanence isolated by AF demagnetization is post-deformational. Analysis of the vec-

tors removed (VR) between successive steps in the same manner indicates that the post-tilting component is the only statistically significant component being removed up to 100 mT.

All specimens that were not step demagnetized were AF cleaned at 30 mT. Uncorrected for tilt, they form a single highly significant mode ($>>99.95\%$) (Fig. 4a). After correction several modes of marginal significance ($\sim 95\%$) (Fig. 4b) appear. Noting that the mode at $D = \sim 280°$, $I = 35°$ represents ENE dipping sites in the Central Belt, and at $D = \sim 90°$,

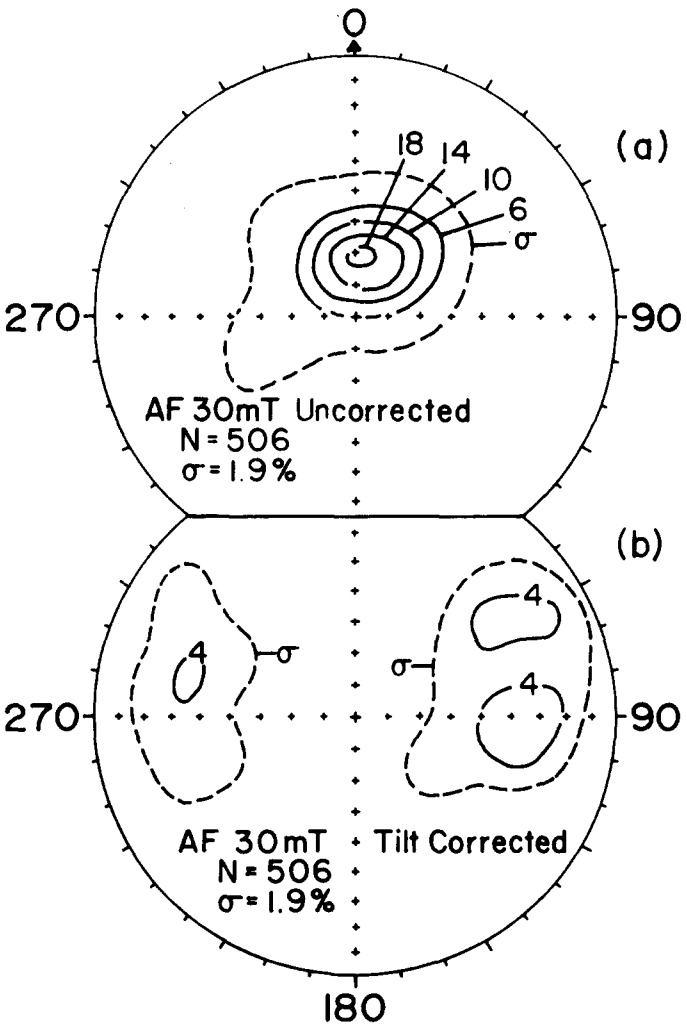


Fig. 4. Population density plots for all 30 mT AF demagnetized directions before (a) and after (b) tilt correction. Conventions as in Fig. 3.

$I = 35°$ represents easterly dipping sites in the Western Belt, the remanence is clearly post-tilting in origin. Therefore, the site mean directions were calculated from the specimen directions without screening or tectonic correction. A fold test using the site means for the Central Belt shows with $>99\%$ confidence that the remanence is post-tilting (Graham, 1949).

THERMAL RESULTS

Of the 56 thermally step demagnetized specimens, 45 isolate a normal polarity vector and 11 a reversed polarity vector. The mean Jn/Jo ratio for the normal specimens (Fig. 5a) decreases nearly linearly with increasing temperature with only a slight suggestion of a "knee" below the 585°C magnetite Curie temperature. The continued decrease of Jn/Jo from ~0.20 above 585°C indicates a substantial hematite component. Similarly the reversed specimens have a small magnetite "knee" below 585°C with a decrease of the intensity ratio from ~0.60 in the hematite temperature above 585°C range. This indicates that hematite is by far the dominant remanence carrier. The remanence directions most typically show ~20° of shift to shallower inclinations between 0 and 400°C on step demagnetization with little change from 400° to 650°C (Fig. 5b, #6 & #25). A few specimens show no directional change at all during thermal cleaning (Fig. 5b, #28) and a very few have very erratic directional changes especially above 565°C (Fig. 5b, #36).

Given the high directional stability of the specimens on thermal cleaning, one specimen per core or two per hand sample were thermally cleaned at 565°C and again at 650°C giving populations of 269 and 268 specimens

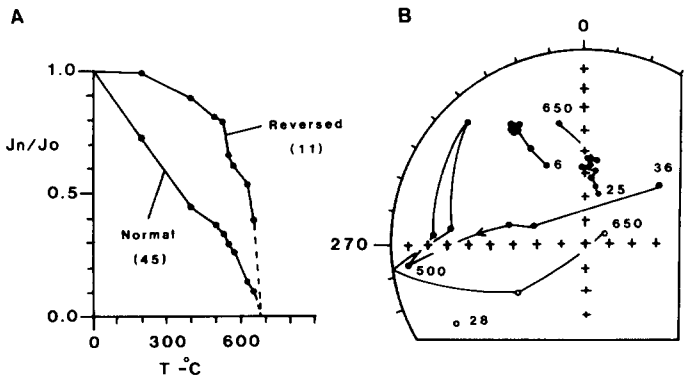


Fig. 5. Thermal step demagnetization data showing: a) remanence intensity decay ratio (Jn/Jo) against temperature (T-C°) for 45 normally and 11 reversely magnetized specimens; and, b) plot of directional changes for four example specimens with the specimen number beside the NRM direction. In (b) note that #28 does not change direction on thermal cleaning.

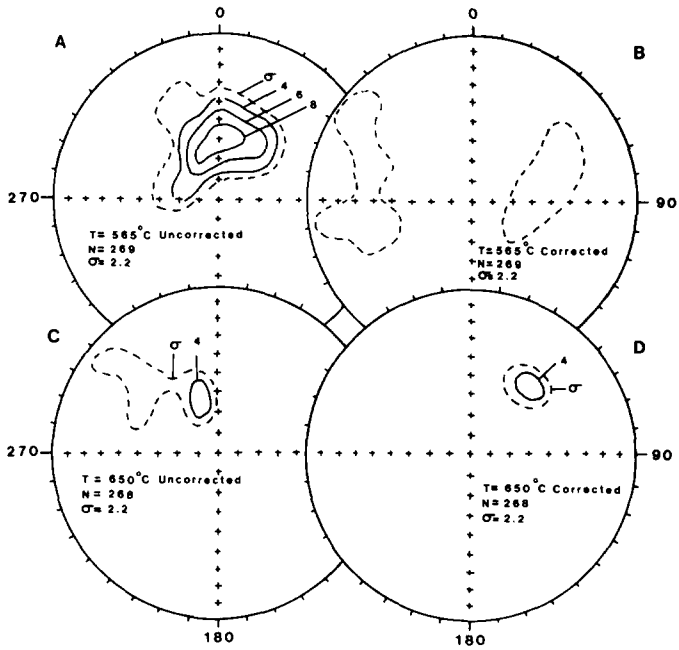


Fig. 6. Population density plots after thermal demagnetization at: a) 565°C before tilt correction; b) 565°C after tilt correction; c) 650°C before tilt correction, and d) 650°C after tilt correction.

respectively after ASD screening. At 565°C the uncorrected directions (Fig. 6a) form a single population of $>>99.95\%$ significance which is dispersed on tilt correction (Fig. 6b) to form two populations of marginal significance ($\sim 95\%$) which reflect the dominant easterly and westerly dips of the Nicola rocks. Again this indicates that the remanence of the B component is post-tilting after 565°C cleaning. At 650°C the uncorrected directions are considerably scattered with two modes of $>95\%$ confidence present. The steeple inclined northerly mode observed at 565°C is still present (Fig. 6c) but its peak intensity is greatly diminished to $\sim 4.1\%$. It is linked by a girdle to a second poorly defined mode with intermediate inclinations in the NW quadrant. On tilt correction a small mode or A component of $\sim 4.3\%$ peak density ($\sim 99.95\%$ significance) is developed in the NE quadrant.

POLISHED SECTIONS

Ten polished sections from a representative selection of flow units were examined. Their metallic mineral content ranges from ~ 0.5 to 5% with an average of $\sim 2\%$. Magnetite occurs as the dominant mineral in four sections and as a trace mineral in two sections. It is found as fine grained euhedra,

anhedra and stringers, and as very fine inclusions in the silicate mineral. The former are extensively corroded, granulated, and altered to hematite. The inclusions are less altered. Two sections show almost total alteration of magnetite to hematite. Minor amounts of pyrrhotite were observed in three sections in association with magnetite. Also trace amounts of titanomagnetite, pyrite, chalcopyrite and ilmenite were seen. These observations confirm that secondary hematite is the dominant stable remanence carrier.

DISCUSSION

The mean directions of the A or 650°C corrected component and of the B or post-tilting component after various demagnetization treatments with their pole positions are given in Table 2.

The A component clearly resides in hematite. It could have been formed by primary deuteric hematization during the Upper Triassic or by predeformational alteration associated with emplacement into the Nicola rocks of several large adjacent plutons including the Lower Jurassic Allison Lake pluton and the Lower Jurassic or younger Penask Batholith. Two arguments favour the Lower Jurassic alternative. First, rift graben tectonics during the Upper Triassic alone are unlikely to have produced the inter-

TABLE II

Mean Remanence Directions and Pole Positions for the Nicola Volcanics

Component, Treatment, Analysis	Data Base		Mean Remanence Direction				Pole Position			
	N	n	R	D°	I°	α_{95}°	Long. °W	Lat. °N	δ_p°	δ_m°
A - tilt corrected										
-650°C, modal*	26	33	25.078	40.7	47.4	5.5	11.3	52.7	4.7	7.2
B - uncorrected										
- 30 mT, tiered	56	496	45.669	0.8	71.4	9.1	116.3	83.9	14.0	16.0
- 565°C, modal*	41	85	38.376	0.4	61.3	5.6	303.0	82.5	6.6	8.6
- 650°C, modal	25	36	24.145	337.0	61.1	5.6	227.6	72.4	6.5	8.5

Notes: N and n indicate the number of sites and specimens respectively used to obtain the means. The mean remanence direction is defined by its vector resultant (R), declination (D°), inclination (I°), and radius of its cone of 95% confidence (α_{95}°). The pole position is defined by its westerly longitude (Long. °W), northerly latitude (Lat. °N), and semiaxies of its oval of 95% confidence (δ_p°, δ_m°). The asterisk denotes the accepted direction.

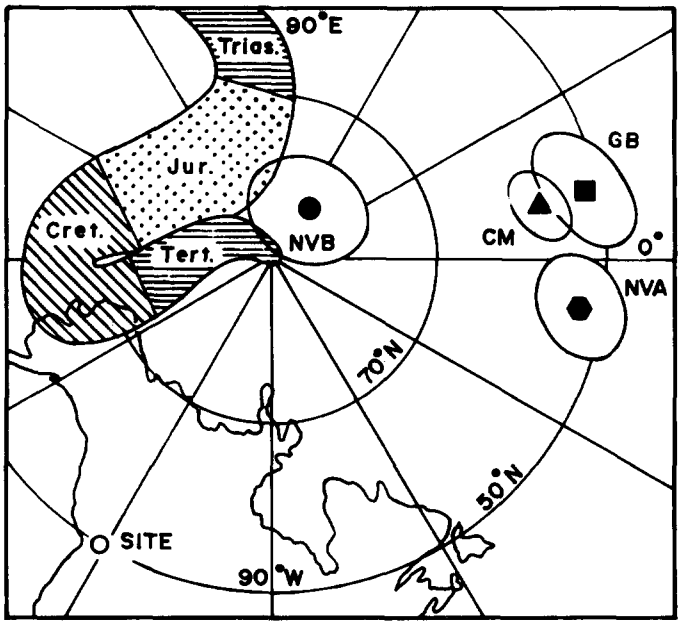


Fig. 7. Polar projection of the northern hemisphere with the APW path of Irving (1979) for cratonic North America showing the locations of the Nicola Volcanics A and B component poles (NVA and NVB) respectively, and the Guichon Batholith (GB) and Copper Mountain Intrusions (CM) poles for Quesnellia with their ovals of 95% confidence.

mediate dips observed. More likely, orogenic activity associated with the Lower Jurassic plutonism was responsible for the monoclinal tilting (Preto, 1979). Second, the pole position for the A component (Fig. 7) is close to those determined for the Lower Jurassic Guichon Batholith (Symons, 1983a) and Copper Mountain Intrusions (Symons and Litalien, 1984), both of which are in the Quesnellia Subterrane just to the north and south of the study area, respectively (Fig. 1). This implies that the Nicola Volcanics have undergone the same post-Early Jurassic clockwise rotation and northward translation history as the Guichon and Copper Mountain plutons. They indicate $43 \pm 7°$ and $39 \pm 6°$ of clockwise rotation and $13 \pm 6°$ and $9 \pm 5°$ of northward translation respectively based on relatively large populations of sites and specimens, and therefore, their results are deemed to be more reliable than the Nicola result of $\sim 55°$ of clockwise rotation and $\sim 5°$ of northward translation based on relatively small populations.

The post-tilting B components is best defined by the 565°C thermal cleaning treatment where a coherent anomaly of high statistical significance involving a large population is isolated (Fig. 6a) and where thermal step demagnetization shows that hematite is the dominant remanence carrier.

The 30 mT AF treatment, based on thermal step demagnetization, retains a hybrid mixture of low blocking temperature ($<500°C$) magnetite and more stable hematite domains (Fig. 5). Also, the lobate pattern of the anomaly isolated after 650°C thermal cleaning (Fig. 6c) indicates the retention of a hybrid mixture of both pre- and post-tilting components.

The age of this post-tilting B component is uncertain. The Lower Cretaceous Kingsvale Group and the Middle Eocene Princeton Group both contain relatively thick basic volcanic sequences that cover much of the study area, and both have been extensively hematized. This suggests that hot hydrothermal solutions of at most Lower Cretaceous age, but more likely of Middle Eocene age were responsible for the hematization of the Nicola rocks. The hematization could also be associated with the extrusion of the flatlying Pleistocene to Recent Valley Basalts which also overlie the Nicola rocks, however, two arguments suggest that this is not likely. Firstly, while extensive, the Valley Basalts contain only five or six basaltic flows with an aggregate thickness of 40 to 50 m (Lambert, 1963) which implies that they represent a short, sharp burst of volcanic activity that is unlikely to have so thoroughly hematized the underlying units. Secondly, the location of the Pleistocene to Recent pole position for cratonic North America is just outside the oval of 95% confidence for the component so that hematization at this time would imply $\sim 7°$ of Recent nortward translation which is both geologically unreasonable (Archibald *et al.*, 1984) and inconsistent with Miocene paleomagnetic data for Terrane I. The Miocene Cariboo Lavas and Cariboo Plugs (Symons, 1969a, b) overly and/or were emplaced into rocks of the Quesnellia, Cache Creek and Stikinia Subterranes. They give concordant pole positions relative to cratonic North America, and therefore show that Terrane I has not undergone statistically significant movement relative to the craton since Miocene time.

The paleomagnetic data from Stikinia provides further constraints on the formation of the B component. The Upper Triassic Takla Group Volcanics, Lower Jurassic Hazelton Group Volcanics, the Lower Jurassic to mid-Cretaceous Axelgold Intrusion (Monger and Irving, 1980) and the Middle to Upper Jurassic Topley Intrusions (Symons, 1983b) all indicate $\sim 14°$ of northward translation relative to the North American craton. Combined with the Upper Triassic to Lower Jurassic paleomagnetic data from Quesnellia, this suggests that Terrane I was assembled adjacent to the craton by Middle Triassic time and that it moved with craton from Upper Triassic to mid-Cretaceous time. If the Nicola Volcanics had been hematized before the mid-Cretaceous, then their B component would indicate a similar rotation and translation to that found for their A component and for the Guichon and Copper Mountain plutons. Correcting the B component for this rotation and translation moves its pole to an

untenable position in the Pacific Ocean off the apparent polar wander (APW) path for the North American craton (Irving, 1979). Therefore, the most likely age for the origin of the post-tilting B component is Middle Eocene, coincident with extrusion of the Princeton Group volcanics. Relative to the Eocene reference pole for the craton, this suggests that $\sim 15°$ of the clockwise rotation and $\sim 7°$ of the northward translation of Quesnellia occurred during Eocene and Oligocene time.

TECTONIC SUMMARY

This paleomagnetic study of the Nicola Volcanics is disappointing in that it is the first volcanic unit in the Canadian Cordillera with very low grade metamorphism to exhibit such extensive postfolding hematization. Nevertheless, its data combined with previous paleomagnetic data provide an initial overview of the geotectonic evolution of Quesnellia and the rest of Terrane I.

By Middle Triassic time, Quesnellia, Stikinia, and the intervening Cache Creek Subterranes were assembled to form the allochthonous composite Terrane I as Monger *et al.* (1982) suggest. This assembly was located adjacent to the western margin of North America about 13° south of its present position, and moved westward with North America as it spread away from the Mid-Atlantic Ridge. Consistent paleolatitudes in Terrane I for the Upper Triassic to mid-Cretaceous time interval signify that it maintained this position until the mid-Cretaceous. This in turn suggests that the Omineca Crystalline Belt to the east of Terrane I, which was formed by the compressional accretion of Terrane I into North America beginning in the Middle Jurassic (Monger *et al.*, 1982; Archibald *et al.*, 1983, 1984), will also show displaced paleolatitudes.

From Upper Cretaceous to Oligocene time, Terrane I moved northward relative to the craton by ~ 1300 km along right-lateral strike slip faults. This is consistent with structural and radiometric evidence given by Monger *et al.* (1982) and Archibald *et al.* (1983, 1984). Paleomagnetic evidence from the Topley Intrusions (Symons, 1983b) indicate that the central interior of Stikinia was not rotated during this event. However, the paleomagnetic data from the Takla Volcanics, Hazelton Volcanics, and Axelgold Intrusion (Monger and Irving, 1980) suggest that minor tectonic blocks on the eastern margin of Stikinia were subjected to substantial clockwise and counterclockwise rotations of more than 100° in some instances, probably as a result of shear between Stikinia and the Omineca Crystalline Belt producing "ball-bearing" type rotations (Beck, 1976). Symons (1971, 1973) summarized the evidence from sedimentary trends in Mesozoic strata and from fault linament trends for the $\sim 45°$ clockwise rotation of, using current ter-

minology, Quesnellia relative to Stikinia. The Nicola A component, Guichon and Copper Mountain paleomagnetic data affirm the sense and magnitude of this rotation relative to the craton, and further suggest that the interior of Quesnellia behaved a coherent subterrane during the rotation. The Nicola B component further indicates that about $\sim 7°$ of the northward translation and about $\sim 15°$ of the clockwise rotation of Quesnellia likely occurred during the mid-Eocene to Oligocene time interval. It is also noteworthy that the clockwise rotation of Quesnellia is consistent in sense and magnitude with those observed in the Cenozoic rocks of Washington and Oregon to the immediate southwest of Quesnellia (e.g., Simpson and Cox, 1977).

The cause of the Upper Cretaceous to Oligocene translation and rotation of Terrane I is not certain. One possibility is that it was associated with the North American plate overriding a plate boundary so that northward motion of the Kula Plate by rifting or of the Pacific Plate by transform faulting (e.g., Atwater, 1970; Stone *et al.*, 1982) was responsible. Another possibility, perhaps interacting with the first, is that oblique subduction of Terrane II to the west with the formation of the intervening Coast Plutonic Complex in Cretaceous to Early Tertiary time (Monger *et al.*, 1982; Archibald *et al.*, 1983, 1984) was responsible.

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